SAM3X / SAM3A Series

Atmel | SMART ARM-based MCU

DATASHEET

Description

The Atmel® | SMART SAM3X/A series is a member of a family of Flash microcontrollers based on the high performance 32-bit ARM® Cortex®-M3 RISC processor. It operates at a maximum speed of 84 MHz and features up to 512 Kbytes of Flash and up to 100 Kbytes of SRAM. The peripheral set includes a High Speed USB Host and Device port with embedded transceiver, an Ethernet MAC, 2 CANs, a High Speed MCI for SDIO/SD/MMC, an External Bus Interface with NAND Flash Controller (NFC), 5 UARTs, 2 TWIs, 4 SPIs, as well as a PWM timer, three 3-channel general-purpose 32-bit timers, a low-power RTC, a low-power RTT, 256-bit General Purpose Backup Registers, a 12-bit ADC and a 12-bit DAC.

The SAM3X/A devices have three software-selectable low-power modes: Sleep, Wait and Backup. In Sleep mode, the processor is stopped while all other functions can be kept running. In Wait mode, all clocks and functions are stopped but some peripherals can be configured to wake up the system based on predefined conditions. In Backup mode, only the RTC, RTT, and wake-up logic are running.

The SAM3X/A series is ready for capacitive touch thanks to the QTouch library, offering an easy way to implement buttons, wheels and sliders.

The SAM3X/A architecture is specifically designed to sustain high-speed data transfers. It includes a multi-layer bus matrix as well as multiple SRAM banks, PDC and DMA channels that enable it to run tasks in parallel and maximize data throughput.

The device operates from 1.62V to 3.6V and is available in 100 and 144-lead LQFP, 100-ball TFBGA and 144-ball LFBGA packages.

The SAM3X/A devices are particularly well suited for networking applications: industrial and home/building automation, gateways.

1. Features

- Core
  - ARM Cortex-M3 revision 2.0 running at up to 84 MHz
  - Memory Protection Unit (MPU)
  - Thumb®-2 instruction set
  - 24-bit SysTick Counter
  - Nested Vector Interrupt Controller
- Memories
  - 256 to 512 Kbytes embedded Flash, 128-bit wide access, memory accelerator, dual bank
  - 32 to 100 Kbytes embedded SRAM with dual banks
  - 16 Kbytes ROM with embedded bootloader routines (UART, USB) and IAP routines
  - Static Memory Controller (SMC): SRAM, NOR, NAND support. NFC with 4 Kbyte RAM buffer and ECC
- System
  - Embedded voltage regulator for single supply operation
  - Power-on-Reset (POR), Brown-out Detector (BOD) and Watchdog for safe reset
  - Quartz or ceramic resonator oscillators: 3 to 20 MHz main and optional low power 32.768 kHz for RTC or device clock
  - High precision 8/12 MHz factory trimmed internal RC oscillator with 4 MHz default frequency for fast device startup
  - Slow Clock Internal RC oscillator as permanent clock for device clock in low-power mode
  - One PLL for device clock and one dedicated PLL for USB 2.0 High Speed Mini Host/Device
  - Temperature Sensor
  - Up to 17 peripheral DMA (PDC) channels and 6-channel central DMA plus dedicated DMA for High-Speed USB Mini Host/Device and Ethernet MAC
- Low-power Modes
  - Sleep, Wait and Backup modes, down to 2.5 µA in Backup mode with RTC, RTT, and GPBR
- Peripherals
  - USB 2.0 Device/Mini Host: 480 Mbps, 4 Kbyte FIFO, up to 10 bidirectional Endpoints, dedicated DMA
  - Up to 4 USARTs (ISO7816, IrDA®, Flow Control, SPI, Manchester and LIN support) and one UART
  - 2 TWI (I2C compatible), up to 6 SPIs, 1 SSC (I2S), 1 HSMCI (SDIO/SD/MMC) with up to 2 slots
  - 9-channel 32-bit Timer Counter (TC) for capture, compare and PWM mode, Quadrature Decoder Logic and 2-bit Gray Up/Down Counter for Stepper Motor
  - Up to 8-channel 16-bit PWM (PWMC) with Complementary Output, Fault Input, 12-bit Dead Time Generator Counter for Motor Control
  - 32-bit low-power Real-time Timer (RTT) and low-power Real-time Clock (RTC) with calendar and alarm features
  - 256-bit General Purpose Backup Registers (GPBR)
  - 16-channel 12-bit 1 msps ADC with differential input mode and programmable gain stage
  - 2-channel 12-bit 1 msps DAC
  - Ethernet MAC 10/100 (EMAC) with dedicated DMA
  - 2 CAN Controllers with 8 Mailboxes
  - True Random Number Generator (TRNG)
  - Register Write Protection
- I/O
  - Up to 103 I/O lines with external interrupt capability (edge or level sensitivity), debouncing, glitch filtering and on-die Series Resistor Termination
  - Up to six 32-bit Parallel Input/Outputs (PIO)

- Packages
  - 100-lead LQFP – 14 x 14 mm, pitch 0.5 mm
  - 100-ball TFBGA – 9 x 9 mm, pitch 0.8 mm
  - 144-lead LQFP – 20 x 20 mm, pitch 0.5 mm
  - 144-ball LFBGA – 10 x 10 mm, pitch 0.8 mm

1.1 Configuration Summary

The SAM3X/A series devices differ in memory sizes, package and features list. Table 1-1 summarizes the configurations.

Table 1-1. Configuration Summary

Feature	SAM3X8E	SAM3X8C	SAM3X4E	SAM3X4C	SAM3A8C	SAM3A4C
Flash	2 x 256 Kbytes	2 x 256 Kbytes	2 x 128 Kbytes	2 x 128 Kbytes	2 x 256 Kbytes	2 x 128 Kbytes
SRAM	64 + 32 Kbytes	64 + 32 Kbytes	32 + 32 Kbytes	32 + 32 Kbytes	64 + 32 Kbytes	32 + 32 Kbytes
NAND Flash Controller (NFC)	Yes	–	Yes	–	–	–
NFC SRAM(1)	4 Kbytes	–	4 Kbytes	–	–	–
Package	LQFP144 LFBGA144	LQFP100 TFBGA100	LQFP144 LFBGA144	LQFP100 TFBGA100	LQFP100 TFBGA100	LQFP100 TFBGA100
Number of PIOs	103	63	103	63	63	63
SHDN Pin	Yes	No	Yes	No	No	No
EMAC	MII/RMII	RMII	MII/RMII	RMII	–	–
External Bus Interface	16-bit data, 8 chip selects, 23-bit address	–	16-bit data, 8 chip selects, 23-bit address	–	–	–
SDRAM Controller(6)	–	–	–	–	–	–
Central DMA	6	4	6	4	4	4
12-bit ADC	16 ch.(2)	16 ch.(2)	16 ch.(2)	16 ch.(2)	16 ch.(2)	16 ch.(2)
12-bit DAC	2 ch.	2 ch.	2 ch.	2 ch.	2 ch.	2 ch.
32-bit Timer	9 ch.(3)	9 ch.(4)	9 ch.(3)	9 ch.(4)	9 ch.(3)	9 ch.(3)
PDC Channels	17	15	17	15	15	15
USART/UART	3/2(5)	3/1	3/2(5)	3/1	3/1	3/1
SPI	1 SPI controller, 4 chip selects + 3 USART with SPI mode	1 SPI controller, 4 chip selects + 3 USART with SPI mode	1 SPI controller, 4 chip selects + 3 USART with SPI mode	1 SPI controller, 4 chip selects + 3 USART with SPI mode	1 SPI controller, 4 chip selects + 3 USART with SPI mode	1 SPI controller, 4 chip selects + 3 USART with SPI mode
HSMCI	1 slot, 8 bits	1 slot, 4 bits	1 slot, 8 bits	1 slot, 4 bits	1 slot, 4 bits	1 slot, 4 bits

Notes:
1. 4 Kbytes RAM buffer of the NFC which can be used by the core if not used by the NFC
2. One channel is reserved for internal temperature sensor
3. Six TC channels are accessible through PIO
4. Three TC channels are accessible through PIO
5. USART3 in UART mode (RXD3 and TXD3 available)
6. Available only on SAM3X8H in LFBGA217 package, which is mounted on SAM3X-EK evaluation kit for SAM3X and SAM3A series. The SAM3X8H device is not commercially available.